

# The Impact of Large Language Model Tutors on Undergraduate Learning

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[Abstract] This thesis investigates the impact of artificial intelligence (AI) tutors on student learning outcomes and teaching assistant (TA) efficiency at a large public university. The research is structured around three primary components. First, office hours data was analyzed from a data science course to assess the need for AI tools. Second, a survey was conducted in a statistics course that utilized a ChatGPT-4o model through Harvard’s PingPong software to gather student feedback on their experiences with AI tutors. Third, these insights informed the development of Edison, an AI-driven tutoring tool. My contribution to Edison, lead by Dr. Narges Norouzi, involved creating a Python-based graphical user interface (GUI), enabling seamless interaction between students and the large language model (LLM). This study employs a mixed-methods approach, integrating data from three key sources: student surveys, office hour queue data, and Edison usage metrics. The primary goal of this research is to demonstrate how AI tutors can enhance office hour efficiency by reducing TA workload and increasing student access to academic support. The findings show that AI tutors can effectively handle simpler questions, provide valuable context to instructors, and decrease wait times for students seeking help. In addition to evaluating the practical benefits of AI tutors, this study also addresses critical ethical considerations, including equity, learning quality, data privacy, and the continued value of human tutoring. By contributing to the evolving landscape of AI in education, this research supports a vision where technology not only promotes personalized learning and accessibility, but also fosters a balanced and thoughtful integration of AI into academic environments. The code referenced in this paper will be posted [here](#).

## 1 Introduction

In a world where information is easily accessible, students are constantly tempted with numerous ways to access answers to their homework. The internet, Chegg, and, more recently, LLMs have made it easier than ever for students to take shortcuts. In reality, these resources are only becoming more prominent. Educators cannot expect students to not use these resources if they lead to better grades and, if used effectively, to a better understanding of the material [4]. The prominence of learning-help platforms combined with lower teacher-to-student ratios compel students to take alternative avenues to receive homework help [2]. In this current era of technological advancements, educators will have to adapt their curriculum to properly respond to these widely available resources.

Instead of letting these students be tempted to quickly obtain answers with little effort, educators can help students make the right choice for their education through course-specific LLM tutors. An LLM tutor is similar to other LLM models (e.g., ChatGPT, Gemini, Claude, etc.), but is trained through various methods to act in greater resemblance to a tutor. While off-the-shelf LLM models may just provide the answer when asked a question by the user, a trained LLM tutor will first ask for the student’s understanding. In theory, trained LLM tutors will steadily guide the student to the solution, rather than providing it to them, encouraging the problem solving process.

In this way, educators are integrating modern technology through encouraging students to use trained LLMs that are approved by the school, over unapproved sources of information. Students will then be motivated to choose the free resource that is introduced and integrated into their course, instead of other online resources. These trained LLMs, or chatbots, should ideally be the only AI resource allowed for that class to discourage the use of other LLMs. The chatbot will, in turn, be more relevant to the course material and will avoid referencing out-of-scope, or incorrect information commonly found on the internet [6][10]. Due to the fact that AI tutors are specifically trained on the course’s material, they will help lead the students to the correct answer similar to how a human tutor would instead of just providing the answer.

Previous work utilized Retrieval-Augmented-Generation (RAG) which provides context to AI-based tutors through relevant data from class materials being dynamically retrieved and integrated into the tutoring process [6][7][11]. RAG and other processes significantly reduce hallucinations and irrelevant rambling by keeping the LLM on topic, helping prevent student confusion [6]. Education chatbots, fine-tuned with methods like RAG, reduce the sourcing of incorrect information resulting in higher accuracy and user satisfaction.

Furthermore, when LLMs are implemented correctly with processes that ensure high-quality responses, studies have documented the positive impact on students. It has been shown to improve time management, and reduce academic uncertainty [9]. It also been demonstrated that LLMs are more consistent than human tutors in their responses [12]. Educational chatbots have also been shown to lead to higher grades and confidence in schoolwork [4]. As demonstrated, AI can be used in a beneficial manner for the student's education, but not without considering notable consequences.

The prominence of AI dependence is only increasing as LLMs become more effective. It has been shown that prolonged AI exposure may lead to dependency and an increase in loneliness [4] [8]. LLMs also lack important human qualities, such as emotional and social awareness, that are better addressed by humans [8]. For example, if a teacher or tutor notices a student getting frustrated and upset while receiving help with homework, the human educator would be able to identify the distress, and help the student emotionally. This emotional intelligence is important to integrate into training models to better provide more empathetic and supportive responses [10].

In addition, AI has a critical weakness for complex tasks such as open-ended mathematical questions which may result in students being mislead in mathematics courses [3]. Similar to mathematics, when AI tutors were utilized and evaluated in physics courses, the LLM's responses consistently ranked lower in both scientific and linguistic quality by students compared to human solutions [5].

One particular example of a chatbot designed to mitigate these common weaknesses is Khanmigo. Khanmigo is developed by Khan Academy and utilizes RAG and prompt engineering to fully train the model as a tutor. The bot is trained to not directly give the answer and ask for the student's reasoning, all while using proper context by referencing the material [7]. In addition, Khanmigo is integrated into Khan Academy's online courses, and is able to cite exact moments in the lesson or video to support the student [7]. This integration and training helps the model not hallucinate, cite correct information, and encourages the student to use class material rather than ask the chatbot for answers. Khanmigo is a significant example of how using an AI chatbot for all education levels can be implemented to support independent learning.

An LLM chatbot was studied in a smaller population at the University of Münster. An LLM-based assistant was integrated into a college robotics course, and was then adjusted to test different techniques such as RAG, fine-tuning, and prompt engineering [6]. Overall, RAG and prompt engineering used in conjunction performed the best across their defined metrics [6]. RAG was shown to increase answer accuracy and reduce hallucination in the model's responses. These characteristics are important for education-centered LLMs to follow, in order to provide high-quality, concise answers for the students who interact with them. This will further encourage the student to utilize school-provided resources as they provide more accurate answers compared to off-the-shelf LLMs.

A similar AI-tutor software, PingPong, developed by Harvard, uses the aforementioned ideals by allowing professors to individually prompt engineer the GPT-4o based system for their class and upload documents containing class material. In turn, the students can access the bot at any time to address questions and their professors can see anonymous chats. PingPong upholds the standard of providing an educational chatbot with the goal of tutoring and providing high quality responses to students. A particular instance of PingPong being used was in the Fall 2024 semester of Statistics 131A at the University of California, Berkeley (UC Berkeley). I surveyed this class to gauge usage, helpfulness, and efficiency of this tool compared to openly available AI tools like Chat-GPT, the results of which will be highlighted later.

Another AI tutor utilized, and developed, at UC Berkeley, is Edison, which is a RAG pipeline based on GPT-4 and trained on teaching assistant (TA) evaluations. Edison was integrated into a class discussion forum, EdStem, in Spring 2024 for Data 100, Data Science 100, or Principles and Techniques of Data Science at UC Berkeley [11]. Unlike PingPong, which allowed students to directly interact with the model, Edison's responses were for the use of TAs to help save time responding to online questions. The TAs could choose whether or not to use Edison's responses, or edit them further. This study yielded results that Edison helped save TA's time when answering content questions on the online forum. The time-savings are significant due to the high office hours wait times highlighted in more depth later in this paper. I created the GUI for Edison so that students will soon be able to directly interact with the model, without the TA approval step.

LLM tutors are often also used as ideas for possible solutions to Benjamin Bloom's "2 Sigma Problem". In Bloom's work as a prominent educational researcher, he concluded that a combination of small group tutoring and mastery learning are the best forms of instruction in terms of student success [1]. The "2 Sigma Problem" is the difficulty of implementing these two forms of instruction while considering funding and faculty restrictions in educational institutions [1]. It is of reasonable assumption that in the future, as LLM tutors improve, AI could be a reasonably affordable solution that supports both one-on-one tutoring and mastery learning.

My contributions are:

- Highlighting limitations of current office hour frameworks in providing timely academic support to students.
- Conducting a survey-based evaluation of PingPong, a new AI-tutoring tool, demonstrating its effectiveness in supported undergraduate coursework.
- Extending Edison to allow for direct student interaction with the model, with the goal of improving education accessibility.

Overall, LLM tutors can help students make better choices and aid their understanding of the material by being a freely available resource that provides high-quality answers to students' questions. By providing key challenges in traditional office hours for large courses, survey data on AI tutoring effectiveness, and enhancing student interaction with AI models like Edison, my work advances the role of artificial intelligence in making undergraduate education more accessible and responsive to student needs.

## 2 Methodology

### 2.1 Overview

This thesis involves analysis of two sources of data in addition to my development of the Edison graphical user interface (GUI). This GUI will serve as the platform in which students interact with the Edison LLM. Prior to the creation of the interface, only TAs were able to utilize Edison in helping to respond to students on the class' discussion forum [11]. The first source of data is from the Data 100 office hours queue from Fall 2023 and Spring 2024, demonstrating how long students wait to receive help in office hours. The second source of data are the survey results concerning usage and opinion of the PingPong chatbot from the Fall 2024 Statistics 131A class offering. Both qualitative and quantitative questions were included, which provided insight on how college students interact and benefit from tutoring bots. The prior data and their corresponding analysis, demonstrate need for Edison to expand to allow for direct student interaction with the model in which the created GUI facilitates. These two sources of data will demonstrate the need, opinion, and outcomes of using two different kinds of LLM tutors in upper division classes at UC Berkeley.

### 2.2 Office Hours Queue

In large undergraduate courses, such as UC Berkeley's Data 100 with nearly 1,500 enrolled students, traditional office hours needed to adapt to effectively support a substantial student population. To address this challenge,

Data 100 implemented an online queue system where students could sign up for help during office hours by specifying the assignment and specific question they were working on. The system then placed students into a queue, organizing them behind earlier requests. Once available, a TA or tutor for the class would find the student, either online or in-person, and provide answers to the student's questions.

Before joining the queue, students were provided with an estimated wait time to receive assistance. Typically, this wait time was under 20 minutes. However, during project weeks, it could surge to over an hour. To analyze this pattern more deeply, I acquired office hours queue data from the Fall 2023 and Spring 2024 semesters of Data 100. This data was sourced from the online queue website provided to me by the lead TA from the course by permission of the professor, Dr. Narges Norouzi. To note, the Fall 2023 data only contained online office hours requests while the Spring 2024 data contained both online and in-person requests. The data was originally stored in a CSV file, which I imported into a Jupyter Notebook and loaded into a Python Pandas dataframe for analysis.

My first step in processing the data was to remove students' email addresses to maintain anonymity. Next, I checked for any rows with null values for either of the "ticket creation" or "ticket update" columns, and there were no null values in this data. To calculate the duration of each help request (or "ticket"), I took the difference from when the ticket was made from when the status was updated to create the column "ticket duration". I then converted the column "ticket duration" into minutes. Additionally, I excluded outliers with a wait time exceeding five hours, as these were likely due to administrative errors, such as a teaching assistant (TA) forgetting to update the online ticket status despite helping the student in person.

I used the Seaborn and Matplotlib libraries to visualize the data. First, I created box plots to visualize the spread of the duration for both semesters for all assignment types. This provided an overall median for the data. Then, I created a histogram that displayed the distribution of ticket durations for only assignments that were projects. Given the high volume of tickets, I filtered the data to focus specifically on project-related tickets, as these represented the highest density of requests. The filtering also assists with graph readability with a legend. The histogram used color-coding to differentiate between the various project types, offering a clear view of how wait times varied across each project.

In the absence of resources to hire more tutors, addressing these extended wait times requires innovative solutions. I used this office hour queue data to explore how AI tutors could help mitigate the strain on traditional office hours in large undergraduate courses. Large Language Models (LLMs) could assist by handling simpler student questions directly or by providing TAs with a summary of a student's interaction with the AI tutor. This approach could enhance efficiency by allowing TAs to focus on more complex queries, ultimately improving the student support experience in large classes.

### 2.3 Survey on the AI-tutor software PingPong

As I began working on the development of the Edison GUI, an AI assistant created at UC Berkeley, I sought to understand the impact of implementing AI chatbots in classroom settings. As others in the Edison research team focused the model's design and accuracy, I was tasked with creating a website for Edison to be reached from. The graphical user interface (GUI) I would later create aimed to facilitate direct interaction between students and the LLM model. Due to my contribution of allowing students to talk directly to the model, it was crucial to study its potential effects in a similar educational environment. Since Edison was set to launch in some of the university's largest undergraduate courses—each with over 1,000 students—it was essential to first analyze a smaller subset of students to identify potential challenges and refine the system before scaling it up.

Fortunately, Statistics 131A, or STAT 131A, Statistical Models for Data Science, was utilizing a tool called PingPong developed by Harvard University. PingPong is a GPT-4o based model designed to be prompted to be a tutor and help university students in their education. PingPong allows for professors to prompt the model and upload course-specific material. Meaning, that PingPong utilizes RAG and prompt engineering to best train

the model to be an effective tutor. In addition, PingPong makes student chats anonymous for the instructor to moderate. The students' names are withheld, but the professor can still view the chats, with the intent of understanding what their students are struggling with, and to ensure the bot is giving valuable feedback. This tool was available for STAT 131A students to use at any time for homework and lab assignments. Dr. Josh Grossman, professor of STAT 131A in Fall 2024, allowed and helped with the creation and launching of the survey.

The use of the AI software PingPong, as well as the similarity in course material to Data 100, made STAT 131A an optimal candidate to survey the usage of education chatbots at UC Berkeley. First, PingPong represents the intended plan for Edison: a chatbot available to students at any time that is specially designed to tutor and is catered towards the course. Edison has been trained to reference course material through a RAG pipeline to better simulate human tutoring [12]. Additionally, both STAT 131A and Data 100 are courses at UC Berkeley that provide an overview on data science principles. Meaning that students in STAT 131A will more or less use the AI tutor in similar ways that Data 100 students would. However, it is notable to mention that STAT 131A is taught in the programming language R while Data 100 is taught in Python using the Pandas library. Overall, due to the similarities between courses, I planned to use the findings from the survey from this smaller course size, 88 students, to better scale Edison for a larger course size, like Data 100.

The survey for the population of STAT 131A students was designed and revised thoroughly with the purpose of gauging PingPong usage, dependence, and opinion. The survey was IRB approved, anonymous, non-incentivized, and online through Google Forms. We restricted each student to only one response each by verifying Berkeley email, but not storing email information. Within the survey were multiple choice, short response, and long response questions, nearly all of which were optional. Both qualitative and quantitative questions were included, which will bring insight on how college students interact and benefit from tutoring bots.

Once ready to receive responses, the survey was set to launch the second to last week of instruction. Dr. Grossman announced the survey in lecture with no incentive other than participating in research of this new tool. As a result, there were 53 responses from a class of 88 students over the course of 2 weeks.

The survey data was saved by Google Forms into a Google Sheet spreadsheet. This data was converted into a CSV file and uploaded into a Python Jupyter notebook. The data was then loaded into a Pandas library dataframe to be analyzed.

### 2.3.1 Survey Questions Grouped by Purpose

The survey was designed to target certain aspects of learning with an AI tutor. In particular, usage, comparison to other LLMs, office hours effect, general feedback, and demographic questions of the users of PingPong. To provide a comprehensive analysis, the survey included a mix of multiple-choice and open-ended questions.

For clarity, the survey questions are grouped below based on their intent, each category addressing a specific aspect of student interaction with AI tutoring. Some questions are used in multiple categories due to their responses possibly fitting both. The following sections detail these question groups and their respective objectives:

#### Usage of PingPong

- 1. How many times have you used PingPong, approximately?
- 2a. Why have you chatted with PingPong less than 5 times?
- 2b. Would you use PingPong more if it were different?

#### PingPong in Comparison to Other LLMs

- 3a. If you wanted to learn more about a statistical concept, would you prefer to use PingPong or another LLM?
- 3b. Please elaborate on your response to the previous question.
- 3e. Do you think PingPong has any notable strengths relative to other LLMs like ChatGPT, Gemini, or Claude?

- 4d. Which of the following would you prefer?
  - Programming + Statistics courses allow students to use any LLM for homework help.
  - Programming + Statistics courses only allow customized chatbots, similar to PingPong.
  - Programming + Statistics courses do not allow students to use any LLM.
- 4e. Please elaborate on your response to the previous question. Why do you prefer your chosen option?

#### LLMs Effect on Office Hours

- 3c. If course staff were not available, what percentage of your questions could PingPong fully answer?
- 4a. Without access to LLMs, I would attend STAT 131A office hours...

#### Strength and Weaknesses of PingPong

- 2b. Would you use PingPong more if it were different?
- 3d. What improvements or changes to PingPong would you like to see implemented?
- 3e. Do you think PingPong has any notable strengths relative to other LLMs like ChatGPT, Gemini, or Claude?

#### Understanding of Class Material

- 4b. Without LLMs, do you think your understanding of the STAT 131A material would be different?
- 4c. Without LLMs, do you think your homework and exam grades in STAT 131A would be different?
- 4f. What concerns, if any, do you have about AI in education? Has PingPong changed your perspective at all?

#### Demographic Questions

- 5a. Which of these classes did you complete before taking STAT131A? Select all that apply
- 5b. What is (are) your major(s)?
- 5c. What is (are) your minor(s)?

The first section, "Usage of PingPong" was meant to assess if students were actually utilizing the software if provided. If students were using the software extensively, that means there is proof of demand for it. If students were not using the software, there were follow-up questions to assess the reasoning of why that was the case.

The second section's, "PingPong in Comparison to Other LLMs", main goal was to assess if students preferred PingPong, an LLM trained to tutor for their class, over an untrained LLM. These questions are to discern if the effort put into trained LLMs to be effective tutors is yielding beneficial results to the student, and hopefully, is encouraging them to choose the trained LLM over the untrained LLM. The question of AI tutors was also extended to other classes, outside of STAT 131A, if the student would like to see similar, trained models in other courses.

Next, I attempt to correlate office hours attendance to AI usage in "LLMs Effect on Office Hours". In this section, the students were challenged to ponder if they would utilize course staff, either through office hours, emails, or online forums, if LLMs were not available. These questions' intentions were to see if LLMs are possibly leading to less office hours usage overall, and their overall helpfulness to the student's education. The purpose was to address fears of AI replacing jobs within education.

To help in the design of the Edison GUI, overall strength and weaknesses of PingPong were queried in the section "Strength and Weaknesses of PingPong". The feedback provided will then be integrated into the design of Edison to better address student's needs. This section also reinforces the distinct differences between untrained LLMs like ChatGPT, Gemini, or Claude, and trained LLMs to see how it benefits the student's learning process.

Furthermore, in "Understanding of Class Material" we wanted to analyze the comprehension of class material and see if LLMs detrimentally removes the problem-solving aspect of education. We challenged the survey respondents to hypothesize what their understanding or grade in the class would be without access to LLMs. Admittedly, this is quite a difficult question to get valid results from, as it is a counterfactual query. In addition,

question 4c would have been a better assessment of AI's effect on grades if split into two questions: one for homework grades and one for exam grades. This would better isolate each due to students being able to use PingPong while working on homework, but not on exams.

Lastly, general demographic questions were asked to possibly find trends between demographic traits and LLM usage.

### 2.3.2 Survey Response

After a two-week period of allowing the survey to accept responses on Google Forms, the number of respondents plateaued. The survey was closed with 53 responses out of a class of 88 students.

The responses were downloaded as a CSV file and uploaded into a Pandas dataframe. From there, each question with multiple choice responses was put into a bar graph. This was done with the Matplotlib Python library. There were some challenges concerning the ordered nature of the responses. For example, in question 1, "Less than 5 chats" is less than "5-10 chats", which has to be manually ordered by the programmer for the bar chart to match.

Particular questions, such as question 4, were further analyzed by splitting each response group by their number of chats. To illustrate, question 4 asked, "If you wanted to learn more about a statistical concept, would you prefer to use PingPong or another LLM?". Each response to this question was then split into the groups established in the first question, or by how many chats they had with PingPong. This was to see if a student uses PingPong more, they may be inclined to prefer PingPong. Demonstrating that if a student is more proficient with the trained LLM, they may find it more beneficial than an untrained LLM.

Questions with short or long responses were read through to analyze student responses. In order to create visualizations for these questions, as I was reading through responses I put them into groups. Often these groups were just "Yes", "No", or "Not Sure". For these questions, I opted for a pie chart visualization since there were fewer groups.

## 2.4 Development of Edison GUI

A portion of this study aimed to develop a graphical user interface (GUI) for an AI-based educational tutor, Edison. Edison is an LLM, with a ChatGPT-4o backend, being developed by researchers at UC Berkeley, with an objective of creating a model that is trained off of TA responses from an online forum, documents from the class curriculum, and by prompt engineering [11][12]. My interface provides a chat window for students to interact with the Edison model.

This interface was developed using Gradio, a Python package for creating machine learning apps. The development process was guided by insights from the previously mentioned PingPong survey. The survey provided valuable data on usability, effectiveness, and desired features, which influenced the design and functionality of the new GUI to optimize student interactions with the AI model.

The survey results indicated that students valued quick, interactive feedback and a user-friendly interface that maintained context over multiple exchanges. Additionally, there was a strong preference for an AI-tutor that facilitated learning without directly providing answers. Students also expressed interest in advanced features such as file upload capabilities, which could enable sharing assignments or notes with the AI.

The development of the GUI occurred over several weeks. During the week of October 21 2024, the initial interface design and development began. The following week, progress was hindered by technical roadblocks related to API integration, outdated Gradio methods, and insufficient documentation. By the week of November 4 2024, a working prototype was completed and presented to the research team, who provided feedback for further refinement. By the week of November 18 2024, the GUI was connected to the Edison API, artificial streaming was set up for the output, and the API endpoint hosting was established.

The GUI utilized Gradio's ChatInterface module to facilitate a chat-based interaction with the AI model. This interface allowed students to engage in a natural, conversational manner, with incremental AI responses simulated

through character-by-character streaming. Survey insights directly informed the implementation of key features, such as maintaining chat history and guiding students toward answers without directly providing them, aligning with the educational goal of promoting guided learning.

A crucial aspect of deploying Edison involved ensuring data privacy and security, particularly concerning API endpoints and chat data storage. The model was hosted on Azure, and a dedicated endpoint was created for the Edison interface. When designing this system, it was essential to consider where student conversations were stored and how data was logged to protect student privacy. Given the educational setting, compliance with FERPA regulations was also a priority, ensuring that no student records were improperly stored or shared. These considerations not only enhanced the security of the Edison interface but also built trust with users, ensuring that their interactions remained confidential and protected.

During development, several challenges were encountered. The Gradio documentation was either overly simplistic or outdated, needing a trial-and-error approach to problem-solving. Additionally, integrating the API required iterative testing to ensure the model responded correctly within the GUI based on chat history. While the initial development goals included enabling PDF upload functionality, this feature was not implemented by the end of the development timeline.

Despite these challenges, the project achieved several successes. The AI model effectively followed instructional prompts, avoiding direct answers to support learning. The GUI's design was enhanced using custom CSS and JavaScript, and the interface maintained conversation history, improving the continuity of tutoring sessions. Overall, the methodology demonstrated how feedback from the PingPong survey guided both the design and development of the AI-tutoring interface, resulting in a functional tool that supports college students' educational needs.

In the future, Edison will be integrated into a larger system that will allow professors to upload their own documents to facilitate a new way for a custom AI tutor to be integrated into many college courses. This expansion will enable a tailored AI tutoring experience, allowing educators to refine the chatbot's responses based on their specific curriculum and teaching styles. Furthermore, additional functionalities, such as enhanced file processing, multi-modal support, and emotional awareness, may be incorporated to improve engagement and comprehension. My contribution, based on findings from the PingPong survey, lays the groundwork for direct student interaction with the model, simulating real-time tutor interactions while maintaining educational integrity and privacy considerations. As Edison evolves, it has the potential to become a fundamental tool in digital education, bridging the gap between students and AI-driven personalized learning.

## 2.5 Discussion

Overall, the data collected from the PingPong survey and the Data 100 office hours queue provides compelling evidence for both the need and effectiveness of AI tutors in higher education, establishing a strong motivation for the Edison project. The PingPong survey highlights a generally positive response from students and demonstrates high engagement with the tool. This suggests that students are not only open to, but enthusiastic about incorporating AI-driven support into their learning processes.

On the other hand, the Data 100 office hours queue data underscores a critical challenge in large classes at UC Berkeley: the need for more timely and effective student support. The extensive wait times, particularly during high-demand periods such as project weeks, reveal a gap in available resources that could be effectively filled by LLM tutors. These AI tutors could reduce wait times by answering straightforward questions and providing valuable context to teaching assistants, thereby optimizing the in-person support that is available.

The Edison Graphical User Interface (GUI) acts as the essential bridge between students and the AI model. By offering an intuitive and accessible platform, the Edison GUI enables students to engage directly with the AI tutor, receiving assistance in real time. This capability is particularly relevant for high-enrollment courses like Data



100, where the volume of student inquiries can quickly overwhelm traditional office hours frameworks. Through Edison, students could receive instant responses to simpler questions, alleviating the burden on teaching staff and enhancing the overall efficiency of the learning environment.

By integrating insights from both the PingPong survey and the Data 100 office hours queue, this project highlights both the demand for and the potential impact of AI-driven educational tools. The positive student engagement with PingPong indicates a readiness to adopt AI support, while the office hours data reveals a pressing need for more scalable assistance in large courses. The Edison GUI directly addresses these challenges by providing an interface for real-time AI tutoring, enhancing both accessibility and efficiency. Together, these findings reinforce the necessity of AI-powered learning solutions and establish Edison as a practical, student-centered approach to improving academic support.

### 3 Results

#### 3.1 Office Hour Queue Data Analysis

The data analysis of the Data 100 office hours queue, divided by project, resulted in two histograms. In the Fall 2023 semester (Figure 1), only online office hours tickets were logged, as demonstrated by the lower count of tickets made. In the Spring 2024 semester (Figure 2), both online and in-person office hours tickets were logged.

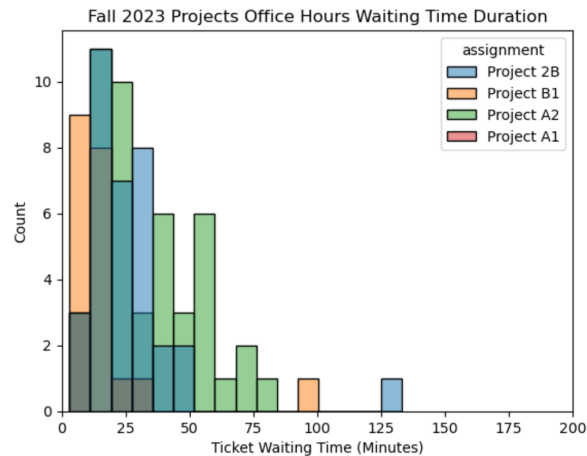


Fig. 1. Fall 2023 Office Hours Queue Data. This histogram displays the distribution of ticket wait times during the Fall 2023 semester, highlighting peak demand periods during project weeks. Note: This semester only contains online office hour tickets.

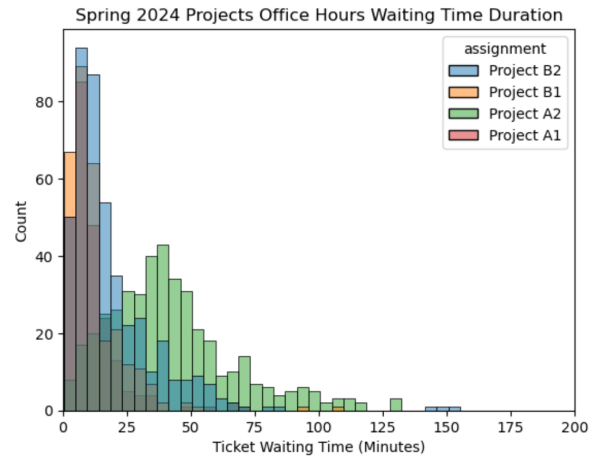


Fig. 2. Spring 2024 Office Hours Queue Data. This histogram displays the distribution of ticket wait times during the Spring 2024 semester, highlighting peak demand periods during project weeks. Note: This semester’s data contains both online and in-person office hours tickets.

The histograms illustrate that a significant proportion of office hour tickets involve waiting times exceeding 50 minutes, with Project A2 showing the longest average waiting time among all projects. While it is encouraging to see that the majority of tickets are resolved within approximately 20 minutes, the prevalence of tickets with extended waiting times is far from negligible. These longer wait times not only indicate periods of high demand but also highlight a bottleneck in the current support system.

These observations indicate a need for innovative solutions, such as AI-driven tutoring systems, to alleviate the pressure on traditional office hours. This technology could help manage the volume of student questions, particularly at large public universities like UC Berkeley, where resources for hiring additional teaching assistants are often limited. By providing immediate support for simpler questions or serving as an initial touch point before students meet with a human tutor, an AI tutor could significantly reduce wait-times and enhance the overall learning experience.

With both human tutoring and LLM tutoring, students can be helped in multiple ways. First, by receiving basic help with the trained LLM, then through follow-up help from the TA. This supports the TA’s workload, while still advocating for the importance of in-person office hours and tutoring.

### 3.2 PingPong Survey Results

The analysis of the survey results will provide valuable insights into student experiences, preferences, and challenges within a course that has AI tutor, PingPong, as a resource. By examining the survey data, I aim to identify key trends and patterns that could inform potential improvements and implementations in AI tutors, like Edison. The following will highlight key results and findings from the survey data.

Question: **1. How many times have you used PingPong, approximately?**

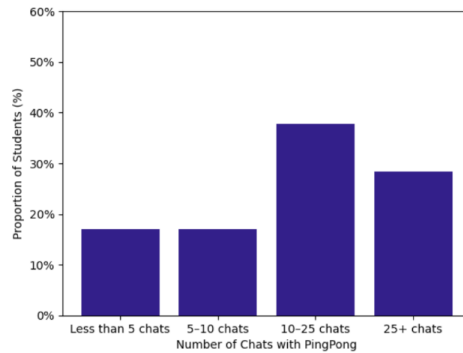


Fig. 3. Distribution of Number of Chats each student had with PingPong.

Above (Figure 3) illustrates that PingPong had high utilization within the STAT 131A class. A majority of the class had over 10 chats. Within a 16 week semester, this could allude to almost weekly usage of PingPong. The demand and usage for AI tutors is evident, with students frequently using the resource.

Question: **2a. Why have you chatted with PingPong less than 5 times?**

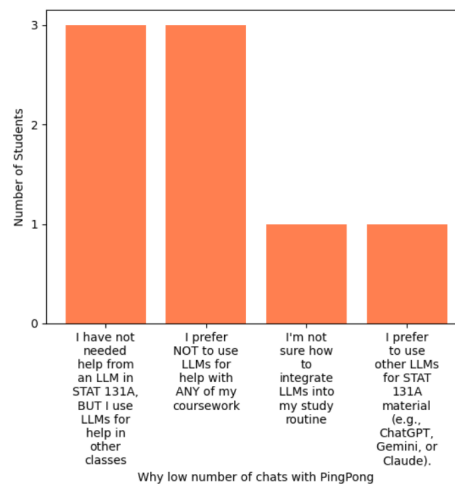


Fig. 4. For students that chose "Less than 5 chats" in question 1, these were their responses.

To understand why some chose to rarely use PingPong, we posed this (Figure 4) question to the student. In addition to their selection in this question, the follow-up question was asked, **"2b. Would you use PingPong more if it were different?"**. Among the changes expressed by students were the following: PingPong feels counterproductive, wished PingPong would cite sources, or that they prefer to not use LLMs in their education.

As PingPong is just an additional class resource, it is understandable and justifiable that some students prefer not to utilize it.

For the students that did use PingPong for a more than 5 chats, we asked the question, "**3d. What improvements or changes to PingPong would you like to see implemented?**". A vast proportion of students, 15 responses out of 31, agreed that the guiding questions are excessive, and, for example, sometimes the query is about simple syntax or definition question that does not need a check of understanding. Another group expressed that PingPong sometimes has difficulties staying on topic and actually answering the student's original question.

Some notable quotes from both 2b and 3d are included below highlighting the weaknesses of PingPong:

- "I find that it tends to make me more confused or take me on a circuitous route."
- "Sometimes it tries to elicit more understanding from you that you don't have and it does not want to proceed in helping you without more of an explanation."
- "For simple/definitions questions, I wish PingPong would print out the answer instead of quizzing us in similar topics."
- "It could be frustrating at times when it was redundant, but I understand that it was necessary."

Question: **3a. If you wanted to learn more about a statistical concept would you prefer to use PingPong or another LLM?**

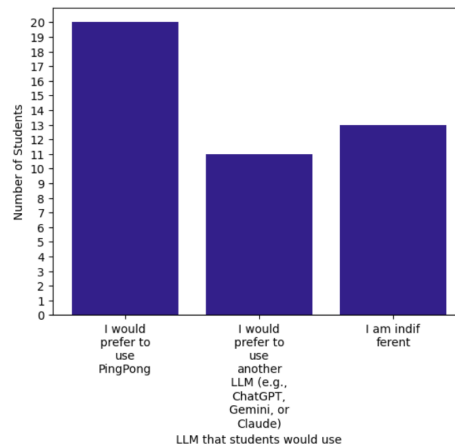


Fig. 5. Student's LLM preferences.

After nearly a semester of using PingPong, this (Figure 5) question was asked to assess whether students preferred it over another LLM. In result, a plurality of students have a preference for PingPong. The findings make it evident that students would opt for a trained LLM over an untrained LLM.

In the following question, students were asked to explain their reasoning. Those who preferred PingPong had one or more of the following reasons: PingPong makes students explain concepts which helps the student figure out the question on their own, and PingPong is more consistent because it is trained for the course specifically. Those who prefer other LLMs say that it is more direct, provides complete answers, and has a better user interface. Those who were indifferent reasoned that both trained LLMs, like PingPong, and untrained LLMs have different use cases and prefer to use both.

Some notable student quotes are highlighted in Table 1 below:

Table 1. Students' LLM Preference for Statistics Learning

| Category          | Sample of Student Quotes                                                                                                                                                                                                                                                                                                                                         |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prefer PingPong   | <p>"I like how PingPong pushes me to remember and explain concepts myself before helping me with my question."</p> <p>"I feel PingPong will actually challenge me to further understand the concept while another LLM such as ChatGPT would just give me the answer."</p>                                                                                        |
| Prefer other LLMs | <p>"I like how PingPong was customized to the class. However, ChatGPT is able to do more."</p> <p>"Other LLMs seem to have more context when I ask questions for studying purposes, as well as the fact that the PingPong model sometimes withholds helpful information in an act to help a student's learning, but I find it unproductive 30% of the time."</p> |
| Indifferent       | <p>"PingPong is better at teaching concepts, other LLMs help you get to answers/quick code fixes faster which are more helpful in certain circumstances."</p>                                                                                                                                                                                                    |

These aforementioned opinions are supported by the responses in a later question, "**3e. Do you think PingPong has any notable strengths relative to other LLMs, like ChatGPT, Gemini, or Claude**". Many of the responses to this question included that the customization of the tool to the course made it easier to use while providing more relevant information. Students also appreciated its resemblance to a tutor, with short responses and guided questions to help the students towards a solution. A representative sample of quotes are as follows, "[PingPong] has knowledge of what our assignments are so it can have more personalized help", "PingPong tends to give more reliable information and breakdowns concepts pretty effectively", "The conversation is more fluid in terms of exchanging knowledge.", and "[PingPong] doesn't give us answers but it helps us understand how to get to the answer."

Question: **3c. If course staff were not available, what percentage of your questions could PingPong fully answer?**

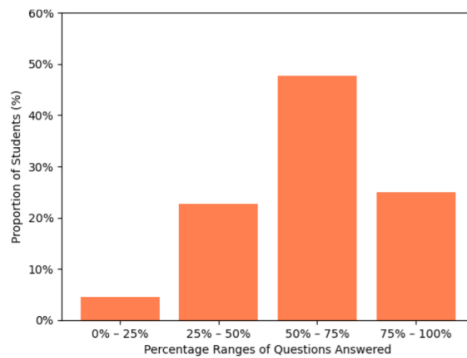


Fig. 6. Approximate percentage of questions answered by PingPong.

### Previous Bar Graph Split into Groups

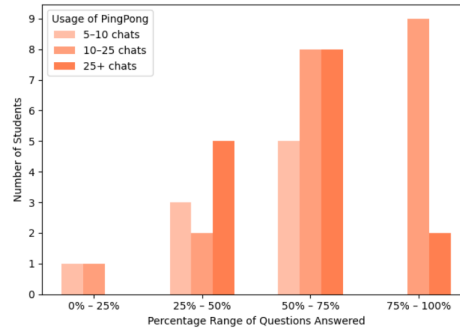


Fig. 7. Approximate percentage of questions answered by PingPong split into usage groups.

The first bar chart (Figure 6) demonstrates that the majority of students report that at least 50% of their questions are answered by PingPong. This leads to considerations such as replacement of course staff, like TAs or graduate student instructors (GSIs), due to a majority of questions being answered. The percentage of questions able to be answered by PingPong, or another LLM tutor, will continue to increase as students better learn how to prompt or ask the model questions and as machine learning advances.

To further discern whether or not better exposure or experience with LLM models leads to more questions answered, I split Figure 6 by the number of chats that students had with the model in Figure 7. There seems to be a slight trend of higher usage with higher amounts of questions answered. However, this is a small sample size, and splitting the graph into this granularity is further splitting that sample; rendering it unreliable for making concrete conclusions.

Question: **4a. Without access to LLMs, I would attend STAT 131A office hours...**

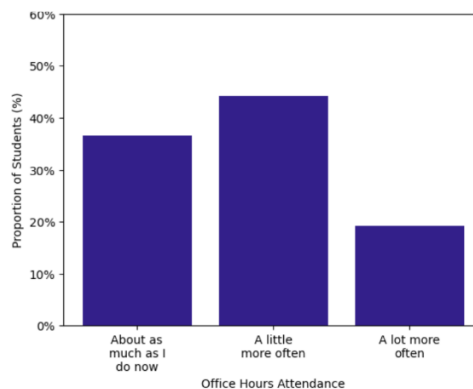


Fig. 8. Distribution of theoretical office hours attendance by students.

According to Figure 8, a plurality of students believed they would attend office hours more frequently if they did not have access to LLMs. This is connected to question 3c (Figure 6), as a majority of students have at least

50% of their questions answered by LLMs.

Question: **4c. Without LLMs, do you think your homework and exam grades in STAT 131A would be different?**

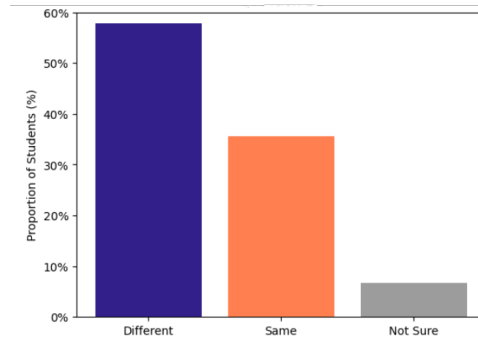


Fig. 9. Chart of hypothetical understanding homework and exam grades.

Although this question (Figure 9) is quite counterfactual, meaning it is difficult to make conclusions of "what if" natured questions, its purpose is to assess whether students believe LLMs benefit or are a detriment to their grades. It is also of note to address that the separation of homework and exam grades into two questions would have better assessed the PingPong's effect. As LLMs are able to be used on homework, but not on exams.

Nevertheless, a plurality of those who responded to this question also reasoned that it is because PingPong is easily accessible in comparison to going to OH or sending an email. This is because PingPong is readily available online, and gives immediate responses. Others who think their grades would be the same, claim they would go to office hours to fill the gaps in their understanding in which PingPong fills.

Many students noted that LLMs make debugging much faster, resulting in less time spent on homework. This can then leave time for studying material or understanding conceptual concepts. In addition, students pointed out that LLMs also help with tedious tasks like formatting documents, summarizing notes, and creating study materials. In this way, the students argue that the time saved leads to better grades.

This eludes to the possibly of administrative and grading tasks being accelerated through the use of AI, to better support TAs and professor to focus on in-person mentoring and tutoring.

This question demonstrates that LLMs have notably effects student's grades, and higher education as a whole, regardless whether it being an off-the-self LLM or trained LLM.

Question: 4d. Which of the following would you prefer?

1. Programming + Statistics courses allow students to use any LLM for homework help.
2. Programming + Statistics courses only allow customized chatbots, similar to PingPong
3. Programming + Statistics courses do not allow students to use any LLM

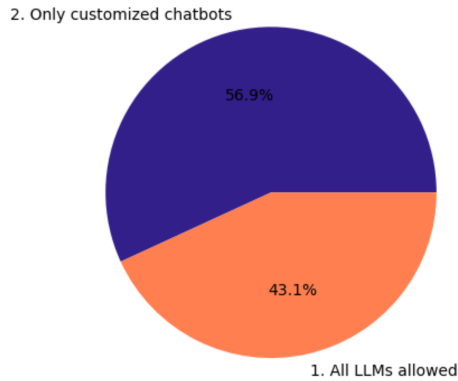


Fig. 10. Pie chart representing preference of LLM usage in future courses.

Nearing the end of the survey, and the end of the semester, this question (Figure 10) urges students to reflect on their LLM usage, and decide what method of AI integration they would prefer for future courses. Notice that not a single student opted for option 3, "Programming + Statistics courses do not allow students to use any LLM".

The next question, **"Please elaborate on your response to the previous question. Why do you prefer your chosen option?"** provided an explanation for the responses. The most common reason for the students who prefer customized chatbots is that they prompt the user for understanding and act as a tutor unlike other LLMs that would just give the answer. Others also say that they favor customized chatbots due to their relevance to the course, their professor's approval, and they provide a method in which students can have access to more educational resources.

Those who prefer if all LLMs were allowed in future courses argue that customized AI tutors like PingPong have certain use cases while other LLMs have theirs, as long as the student is honest about their usage. A small proportion of students believe it is important for students to learn how to navigate unrestricted LLM use on their own, as the non-academic world will not contain "guardrails" like PingPong.

A representative sample of students' quotes are collected below in Table 2 :

Table 2. Students' LLM Preference for Future Courses

| Category                | Sample of Student Quotes                                                                                                                                                                                                                         |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customized LLMs Allowed | <p>"Any LLM is just harmful for student's learning, but I feel like I am able to leave PingPong actually understanding concepts and material."</p> <p>"I like when I know the instructor can customize and suggest an LLM that I can trust."</p> |
| All LLMs Allowed        | <p>"I think people revert to GPT regardless"</p> <p>"People study in different ways and other LLMs might cater to that more."</p>                                                                                                                |



Question: **4f. What concerns, if any, do you have about AI in education? Has PingPong changed your perspective at all?**

This question was a free response question, with the purpose of gauging if students shared common concerns about the prominence of AI with educators. Notable responses include:

- *"I'm worried that LLMs will be used in place of human tutors."*
- *"I'm concerned about students' understanding of material because of overuse of LLMs. I think that something like PingPong is better than other LLMs, although it does still allow for a certain amount of copy-and-pasting that can bypass some of the learning process."*
- *"While I like how AI has helped in making tasks more efficient, I prefer the time when people put in the work and had to put thought into their work. I fear AI is making us lose our creativity and original thoughts."*
- *"I hope instead of banning the use of LLMs in classrooms, we should be taught to integrate it into our work. I like how we were taught to use Ping Pong as a tool in this course."*
- *"PingPong has made me trust in AI and believe in the potential of AI more in terms of aiding my understanding of material and education."*
- *"I am optimistic about the future of AI in education. I think it's a tool that can help students learn at their speed with less fear of asking questions."*

### 3.3 Edison GUI

Just as PingPong was a method of which students could directly interact with a model to receive homework help without immediately being given the answer, the Edison GUI makes it possible for students to do the same with the Edison LLM. To re-iterate, Edison is a ChatGPT-4o model with retrieval-augmented generation from course materials [12][11]. These enhancements enable more context-aware responses by integrating course-specific resources.

Overall, this (Figure 11) is how the interface appears to the user when first opened. In addition, the top banner appears letter by letter to simulate how an LLM responds to students.

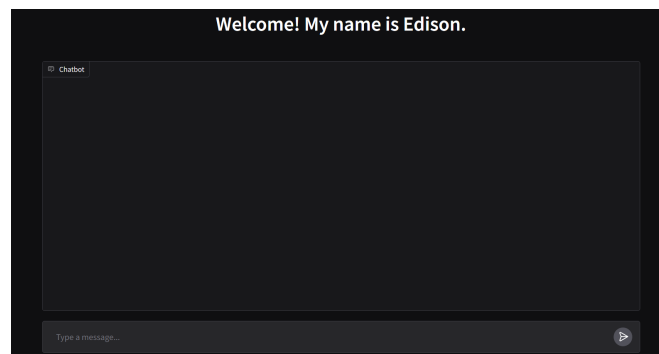


Fig. 11. How the chat window appears to the student

In addition here (Figure 12) is an example of a question a student may have in Data 100:

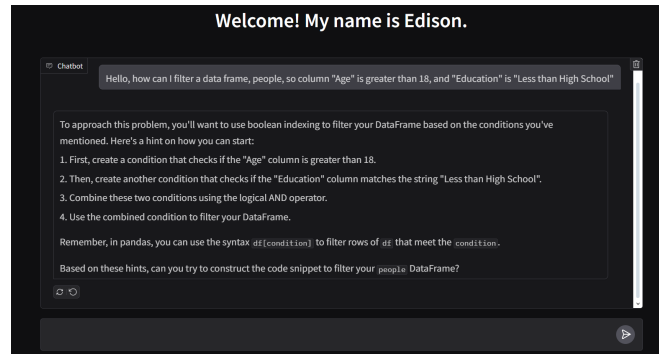


Fig. 12. First Example

Another Example (Figure 13):

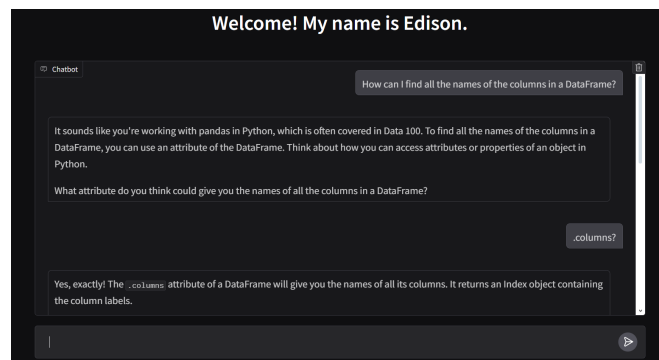


Fig. 13. Second Example

This interface will, in the near future, be integrated as a resource within the Data 100 course, and possibly other courses as an additional resource for students to utilize. The Edison GUI facilitates a method in which students can interact with large language models in a way that better benefits their learning process.

In addition, the GUI was tested by current Data 100 TAs who suggested that the character streaming is a good speed for a conceptual question, but may benefit from being faster for debugging questions. This aspect of the GUI will be improved upon later to determine which speed would be more optimal, as it is unrealistic to adjust output speed based on question type. They also want to see saved conversations and document upload in future updates of the Edison GUI.

### 3.4 Discussion of Results

Overall, a substantial proportion of Data 100 office hours tickets are addressed in under 20 minutes, which is exceptional due to the class having an over 1000 student class size. However, 15% of student requests take over an hour to be addressed which is a significant amount of time for the student to wait before getting assistance.

During this time, students may keep trying to figure out the problem, or they might become discouraged, office hours could end, or they may have another commitment to attend before receiving help.

Long wait times discourage students and decrease access to educational resources. Hence, there is a need for additional resources, like LLM tutors, to help students. Educational chatbots cannot answer every question, as seen in the PingPong survey, but can answer a majority while leaving complex questions for the TA or professor to help with. This additional resource can help support long wait times in office hours.

Furthermore, chatting with the AI tutor can lead the student to the answer on their own or provide more context for both the student and human tutor to understand what that student is struggling with. For the student, the AI tutor will ask guiding questions to lead the student to the right answer. Resulting in a better understanding of the material, and possibly, a more coherent question for the TA to answer. For the TA, they can use the chat history to understand what the student is asking.

This is all not to suggest that students should primarily rely on LLMs for help with assignments, students should have full agency on how they would like their questions to be answered. Educational AI tutors provide another mode of homework help for students.

For the students that do choose to use tutor-trained chatbots, however, they generally have a positive opinion as seen in the PingPong survey results. When provided within the course, majority of students use the resource and appreciate that it is specialized towards the course, approved by the professor, is constantly available, and aims to the tutor the user.

However, PingPong also received its fair share of criticism. Some students felt that it does not fully act as a tutor because it keeps prompting for understanding, it is not able to identify when the student simply needs more context. Students also felt that the trained LLM went in circles and was redundant. Trained LLMs, like PingPong and Edison, are extremely hard to train to produce outcomes that balance withholding the answer from the student, while providing related information to help with problem solving.

This is one of many ways AI tutors are simply not equivalent to human tutors. Trained LLM tutors find it difficult to include relevant examples without explicitly answering the question, and are unable to identify when the student may begin to feel frustrated. On the other hand, these are both traits that human tutors excel in.

These positive attributes of PingPong, as well as the negative ones, were considered while developing Edison. As the Edison model was being developed by others on the Edison research team, I created an interface which will allow students to directly interact with the model to receive tutoring help. Nevertheless, this GUI needs further development which includes: determining output speed, adding file upload, and saving student conversation in a FERPA-compliant manner.

### 3.4.1 *Ethical Considerations in Encouraging an AI-Powered Education*

In my research, it can be argued that developing this technology inherently discourages human tutoring systems, enables students to skip vital problem-solving processes, and encourages AI usage in education. However, while is to be emphasized that the value of human interaction for tutoring is irreplaceable, and that this technology is designed to help human tutors.

Resulting from this optimization, more students will be able to interact with tutors for the complex questions they may have, as well as reduce workload and stress on the TA. From the Data 100 office hour queue data I analyzed before, I discovered that for approximately 15% of tickets, the tutor takes over an hour to get to the student. These immensely long wait times are detrimental to the student's learning process, and will be utilized to demonstrate the help that TAs need. Unless the university allocates more funding to hire and train more TAs, the best response to this issue is implementing the LLM tutor.

In addition, technologies, like PingPong or Edison, are reactionary. They are designed to discourage usage of off-the-shelf LLMs that completely discard the problem solving aspect of learning for the student. By providing

a resource that better benefits student's education, these technologies discourage shortcuts offered by widely-available LLMs. Students will continue to use AI as long as it is open to all. Therefore, it is essential for educators to guide students toward making responsible and productive choices by offering AI tools that genuinely support their learning, like LLM tutors. When educators prioritize making AI resources more accessible and effective, students are more likely to use these tools in ways that enhance their education due to professor's support and recommendation as seen in feedback within the PingPong survey.

Additionally, common inaccuracies in AI responses can negatively affect students, but this is unavoidable and occurs in human tutoring as well. However, methods such as RAG, fine-tuning, and prompt engineering help increase accuracy and relevance of the model's responses. As a whole, it is better to get any help, even if a small portion is incorrect, than to receive no help at all. Through advancements in machine learning, the accuracy of responses by models will continue to improve.

Another implication of this technology is allowing constant access to tutoring resources. This attribute opens tutoring to those with disabilities who may have difficulties getting to or learning from an in-person office hours. For individuals who find it difficult to get in-person help, this technology makes educational help equitable, and enables them to get access to tutoring resources.

Furthermore, this technology can be provided to schools that do not have the funding for substantial tutoring resources as an affordable alternative. This will encourage equitable access to education across the world, and in lower income communities. As previously mentioned, it is a possible partial solution to Bloom's "2 Sigma Problem" that facilitates small-group tutoring and mastery learning if AI tutors are implemented correctly [1].

In addition, students may be afraid to ask human tutors questions due to embarrassment or lack of understanding of the material. In contrast, students may feel more willing to ask an LLM more simple questions allowing them to catch up on class content. This can then enable these students to better ask questions or participate in class, and prevent a build-up of misunderstanding of the course so far.

The final ethical component I will address is data privacy. The PingPong chatbot is anonymous, but allows for the instructor to view chats. This will protect the student's identity, but allow for the instructor to moderate usage and understand which topics their students need the most help with. Edison keeps student's chats with the model private, which is essential for creating a judgment-free resource, and ensures that the model that does not train on student conversations with it.

While the implementation of AI tutoring systems presents ethical challenges and potential drawbacks, it also offers significant opportunities to support human tutoring, reduce wait times, and expand access to educational support. By addressing issues such as data privacy, equitable access, and the trade-off between human and LLM tutors, this technology can be utilized as a tool that enhances, not replaces, traditional educational methods.

## 4 Conclusion

The implementation of AI tutoring systems presents both challenges and opportunities in education. While concerns exist regarding their impact on human tutoring, problem-solving skills, and ethical considerations like data privacy, AI tutors can significantly enhance student support. By reducing long wait times, providing continuous access to help, and encouraging responsible AI use, tools like PingPong and Edison serve as valuable supplements to traditional tutoring.

In the future, PingPong and Edison will be implemented in more courses and will be improved upon. In particular, Edison will allow for file upload and save conversation for the student's reference.

These technologies ensure that students receive guidance tailored to their needs while maintaining human tutors for complex inquiries. Ultimately, AI-powered tutors should be viewed not as replacements, but as enhancements to existing educational frameworks, fostering a more accessible and effective learning environment.

## Acknowledgments

To Professor Raupach and Professor O’Connell, for showing me that education is for everyone.

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## 5 Appendices

### A Reproducibility Documentation

Below details how to replicate the work done in this thesis on your local computer. The files are at this Github repository: <https://github.com/hexhannah/data-sci-honors-thesis/tree/main>

#### A.1 PingPong Survey Analysis

To begin, put the `SurveyAnalysis.ipynb` file and a data folder in the same directory of a Python programming workspace. I personally used a Jupyter Notebook. Within the data folder, I inserted the `PingPong_SurveyResponses.csv`. Make sure to adjust the file name according to your setup when calling pandas to read the CSV file.

The rest of the notebook should run smoothly, if setup was completed successfully.

## A.2 Edison GUI

First, put the `Ed_Interface_File.ipynb` file into a Python programming workspace. I used a Jupyter Notebook. As is, this is just code for a front end interface. It is necessary to add an API Key and endpoint URL at the start of the code file. It will also not function as the LLM Edison without its specific API. However, this code can be used with any LLM API and endpoint setup as a front end website. With just the provided code, it is locally hosted with a public link for a limited time period.

## B Full Survey

### Information given to students at start of survey:

"This voluntary survey collects student opinions on large language models (LLMs) to help inform an LLM framework for use in UC Berkeley classes.

This Google Form anonymously verifies that you are a UC Berkeley affiliate. Your UC Berkeley email address and contact information are not visible to the researchers.

As with any survey, confidentiality cannot be fully guaranteed. Please note that any question may be skipped. Anonymized responses will be publicly available at the UC Berkeley Library.

An aggregated survey summary will be shared with all respondents.

This survey is being conducted by an undergraduate in the data science honors program as part of their honors thesis.

Contact for questions: [hannaheb@berkeley.edu](mailto:hannaheb@berkeley.edu)

CPHS Protocol ID: 2024-10-17848

Background Information: LLM (Large Language Model): An AI (Artificial Intelligence) that can understand and generate text like ChatGPT, Gemini, or Claude.

Estimated completion time: Approximately 5–10 minutes."

### (1) Frequency of PingPong Use How many times have you used PingPong, approximately?

- 25+ chats
- 10-25 chats
- 5-10 chats
- Less than 5 chats

### (2) If "Less than 5 chats" selected in Question 1

If the respondent did select "Less than 5 chats" in Question 1, they were asked additional questions. These questions included:

#### (a) Why have you chatted with PingPong less than 5 times?

- I prefer to use other LLMs for STAT131A material (e.g., ChatGPT, Gemini, or Claude)
- I have not needed help from an LLM in STAT 131A, but I use LLMs for help in other classes
- I prefer NOT to use LLMs for help with ANY of my coursework
- I'm not sure how to integrate LLMs into my study routine

#### (b) Would you use PingPong more if it were different?

*If not, why not? If yes, what would need to change about PingPong for you to use it more?*

- Long Answer Response

### (3) More than 5 chats from selection in Question 1

If the respondent did not select "Less than 5 chats" in Question 1, they were asked additional questions. These questions included:

#### (a) If you wanted to learn more about a statistical concept, would you prefer to use PingPong or another LLM?

- I would prefer to use PingPong

- I would prefer to use another LLM (e.g., ChatGPT, Gemini, or Claude)
  - I am indifferent
- (b) **Please elaborate on your response to the previous question.** *If you prefer one LLM over another, why? If you are indifferent why?*
- Long Answer Response
- (c) **If course staff were not available, what percentage of your questions could PingPong fully answer?**
- 0%-25%
  - 25%-50%
  - 50%-75%
  - 75%-100%
- (d) **What improvements or changes to PingPong would you like to see implemented?** *For example, do you find any features of PingPong frustrating or confusing? If yes, how would you improve those features?*
- Long Answer Response
- (e) **Do you think PingPong has any notable strengths relative to other LLMs like ChatGPT, Gemini, or Claude?** *If yes, which features do you think are the most impactful?*
- Long Answer Response
- (4) **All students regardless of PingPong usage**
- These questions were posed to all students regardless of PingPong usage
- (a) **Without access to LLMs, I would attend STAT 131A office hours...**
- About as much as I do now
  - A little more often
  - A lot more often
- (b) **Without LLMs, do you think your understanding of the STAT 131A material would be different?**
- If yes, why do you think your understanding would be different? If not, why do you think your understanding would be the same?*
- Long Answer Response
- (c) **Without LLMs, do you think your homework and exam grades in STAT 131A would be different?**
- If yes, why do you think your grades would be different? If not, why do you think your grades would be the same?*
- Long Answer Response
- (d) **Which of the following would you prefer?**
1. Programming + Statistics courses allow students to use any LLM for homework help.
  2. Programming + Statistics courses only allow customized chatbots, similar to PingPong
  3. Programming + Statistics courses do not allow students to use any LLM
- 1. All LLMs allowed
  - 2. Only customized chatbots
  - 3. No LLMs at all
- (e) **Please elaborate on your response to the previous question. Why do you prefer your chosen option?**
- Long Answer Response

- (f) **What concerns, if any, do you have about AI in education? Has PingPong changed your perspective at all?** *For example, are you scared that LLMs may replace human tutors, interpersonal interaction, or office hours? On the other hand, are you optimistic about the future of AI in education?*

- Long Answer Response

- (5) **Demographic Questions** These questions were asked of all students.

- (a) **Which of these classes did you complete before taking STAT131A? Select all that apply**

- Data 8
- Stat 20
- Data 88S (or other Probability Theory)
- CS61A/DataC88C (or other Programming)
- Stat 133
- None of the above

- (b) **What is (are) your major(s)?** *Please separate majors with commas.*

- Short Answer Response

- (c) **What is (are) your minor(s)?** *Please separate minors with commas.*

- Short Answer Response

END OF SURVEY